Installing packages

```
# pip install numpy
# pip install pandas
# pip install matplotlib
# pip install seaborn
```

Importing packages

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Reading and Writng data

```
df = pd.read_csv("student_scores.csv")
print(df.head())
```

```
   Unnamed: 0  Gender EthnicGroup          ParentEduc     LunchType
TestPrep  \
0           0  female         NaN   bachelor's degree      standard
none
1           1  female     group C        some college      standard
NaN
2           2  female     group B     master's degree      standard
none
3           3    male     group A  associate's degree  free/reduced
none
4           4    male     group C        some college      standard
none

  ParentMaritalStatus PracticeSport IsFirstChild  NrSiblings
TransportMeans  \
0             married     regularly          yes         3.0
school_bus
1             married     sometimes          yes         0.0
NaN
2              single     sometimes          yes         4.0
school_bus
3             married         never           no         1.0
NaN
4             married     sometimes          yes         0.0
school_bus

  WklyStudyHours  MathScore  ReadingScore  WritingScore
0            < 5         71            71            74
1         5 - 10         69            90            88
2            < 5         87            93            91
```

```
3          5 - 10          45             56             42
4          5 - 10          76             78             75
```

```
df.describe()
```

```
         Unnamed: 0    NrSiblings     MathScore  ReadingScore
WritingScore
count  30641.000000  29069.000000  30641.000000  30641.000000
30641.000000
mean     499.556607      2.145894     66.558402     69.377533
68.418622
std      288.747894      1.458242     15.361616     14.758952
15.443525
min        0.000000      0.000000      0.000000     10.000000
4.000000
25%      249.000000      1.000000     56.000000     59.000000
58.000000
50%      500.000000      2.000000     67.000000     70.000000
69.000000
75%      750.000000      3.000000     78.000000     80.000000
79.000000
max      999.000000      7.000000    100.000000    100.000000
100.000000
```

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 30641 entries, 0 to 30640
Data columns (total 15 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Unnamed: 0           30641 non-null  int64
 1   Gender               30641 non-null  str
 2   EthnicGroup          28801 non-null  str
 3   ParentEduc           28796 non-null  str
 4   LunchType            30641 non-null  str
 5   TestPrep             28811 non-null  str
 6   ParentMaritalStatus  29451 non-null  str
 7   PracticeSport        30010 non-null  str
 8   IsFirstChild         29737 non-null  str
 9   NrSiblings           29069 non-null  float64
 10  TransportMeans       27507 non-null  str
 11  WklyStudyHours       29686 non-null  str
 12  MathScore            30641 non-null  int64
 13  ReadingScore         30641 non-null  int64
 14  WritingScore         30641 non-null  int64
dtypes: float64(1), int64(4), str(10)
memory usage: 3.5 MB
```

```
df.head(10)
```

```
   Unnamed: 0  Gender EthnicGroup          ParentEduc     LunchType  \
0           0  female         NaN   bachelor's degree      standard
1           1  female     group C        some college      standard
2           2  female     group B     master's degree      standard
3           3    male     group A  associate's degree  free/reduced
4           4    male     group C        some college      standard
5           5  female     group B  associate's degree      standard
6           6  female     group B        some college      standard
7           7    male     group B        some college  free/reduced
8           8    male     group D         high school  free/reduced
9           9  female     group B         high school  free/reduced

    TestPrep ParentMaritalStatus PracticeSport IsFirstChild
NrSiblings  \
0       none             married     regularly          yes
3.0
1        NaN             married     sometimes          yes
0.0
2       none              single     sometimes          yes
4.0
3       none             married         never           no
1.0
4       none             married     sometimes          yes
0.0
5       none             married     regularly          yes
1.0
6  completed             widowed         never           no
1.0
7       none             married     sometimes          yes
1.0
8  completed              single     sometimes           no
3.0
9       none             married     regularly          yes
NaN

  TransportMeans WklyStudyHours  MathScore  ReadingScore  WritingScore

0     school_bus            < 5         71            71            74

1            NaN         5 - 10         69            90            88

2     school_bus            < 5         87            93            91

3            NaN         5 - 10         45            56            42

4     school_bus         5 - 10         76            78            75

5     school_bus         5 - 10         73            84            79

6        private         5 - 10         85            93            89
```

```
7        private           > 10         41            43            39

8        private           > 10         65            64            68

9        private            < 5         37            59            50
```

```
df.isnull().sum()
```

```
Unnamed: 0                0
Gender                    0
EthnicGroup            1840
ParentEduc             1845
LunchType                 0
TestPrep               1830
ParentMaritalStatus    1190
PracticeSport           631
IsFirstChild            904
NrSiblings             1572
TransportMeans         3134
WklyStudyHours          955
MathScore                 0
ReadingScore              0
WritingScore              0
dtype: int64
```

Drop (delete) "unnamed" column

```
df = df.drop("Unnamed: 0", axis=1)
df
```

```
       Gender EthnicGroup          ParentEduc     LunchType   TestPrep
\
0      female         NaN   bachelor's degree      standard       none

1      female     group C        some college      standard        NaN

2      female     group B     master's degree      standard       none

3        male     group A  associate's degree  free/reduced       none

4        male     group C        some college      standard       none

...       ...         ...                  ...           ...        ...

30636  female     group D         high school      standard       none

30637    male     group E         high school      standard       none

30638  female         NaN         high school  free/reduced  completed
```

```
30639  female     group D  associate's degree      standard  completed

30640    male     group B        some college      standard       none


      ParentMaritalStatus PracticeSport IsFirstChild  NrSiblings  \
0                 married     regularly          yes         3.0
1                 married     sometimes          yes         0.0
2                  single     sometimes          yes         4.0
3                 married         never           no         1.0
4                 married     sometimes          yes         0.0
...                   ...           ...          ...         ...
30636              single     sometimes           no         2.0
30637              single     regularly           no         1.0
30638             married     sometimes           no         1.0
30639             married     regularly           no         3.0
30640             married         never           no         1.0

      TransportMeans WklyStudyHours  MathScore  ReadingScore
WritingScore
0         school_bus            < 5         71            71
74
1                NaN         5 - 10         69            90
88
2         school_bus            < 5         87            93
91
3                NaN         5 - 10         45            56
42
4         school_bus         5 - 10         76            78
75
...              ...            ...        ...           ...
...
30636     school_bus         5 - 10         59            61
65
30637        private         5 - 10         58            53
51
30638        private         5 - 10         61            70
67
30639     school_bus         5 - 10         82            90
93
30640     school_bus         5 - 10         64            60
58

[30641 rows x 14 columns]
```

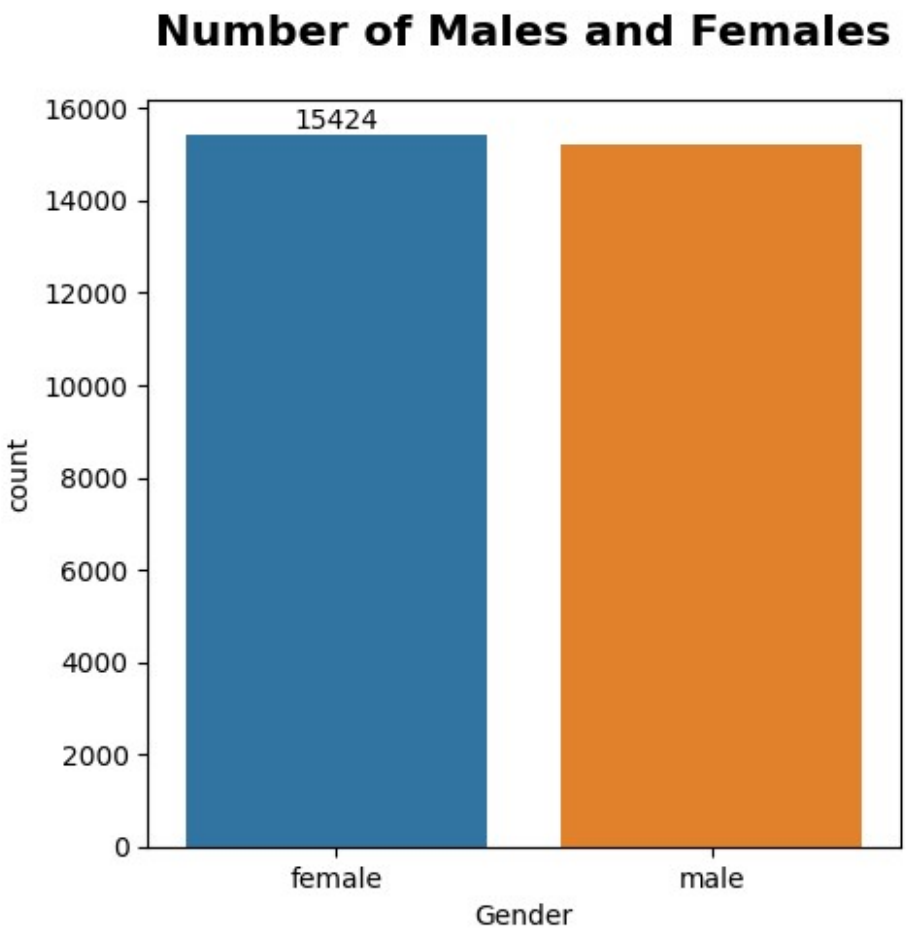
Change weekly study hours column

```
df["WklyStudyHours"] = df["WklyStudyHours"].str.replace("05-Oct","5-
10")

df.head()
```

```
   Gender EthnicGroup          ParentEduc     LunchType TestPrep  \
0  female         NaN   bachelor's degree      standard     none
1  female     group C        some college      standard      NaN
2  female     group B     master's degree      standard     none
3    male     group A  associate's degree  free/reduced     none
4    male     group C        some college      standard     none

  ParentMaritalStatus PracticeSport IsFirstChild  NrSiblings
TransportMeans  \
0             married     regularly          yes         3.0
school_bus
1             married     sometimes          yes         0.0
NaN
2              single     sometimes          yes         4.0
school_bus
3             married         never           no         1.0
NaN
4             married     sometimes          yes         0.0
school_bus

  WklyStudyHours  MathScore  ReadingScore  WritingScore
0            < 5         71            71            74
1         5 - 10         69            90            88
2            < 5         87            93            91
3         5 - 10         45            56            42
4         5 - 10         76            78            75
```

Gender Distribution

```
plt.figure(figsize=(5,5))
ax= sns.countplot(data= df , x="Gender", hue="Gender")
ax.bar_label(ax.containers[0])
plt.title("Number of Males and Females", fontweight='bold', fontsize=
16 , pad= 20)
plt.show()
```

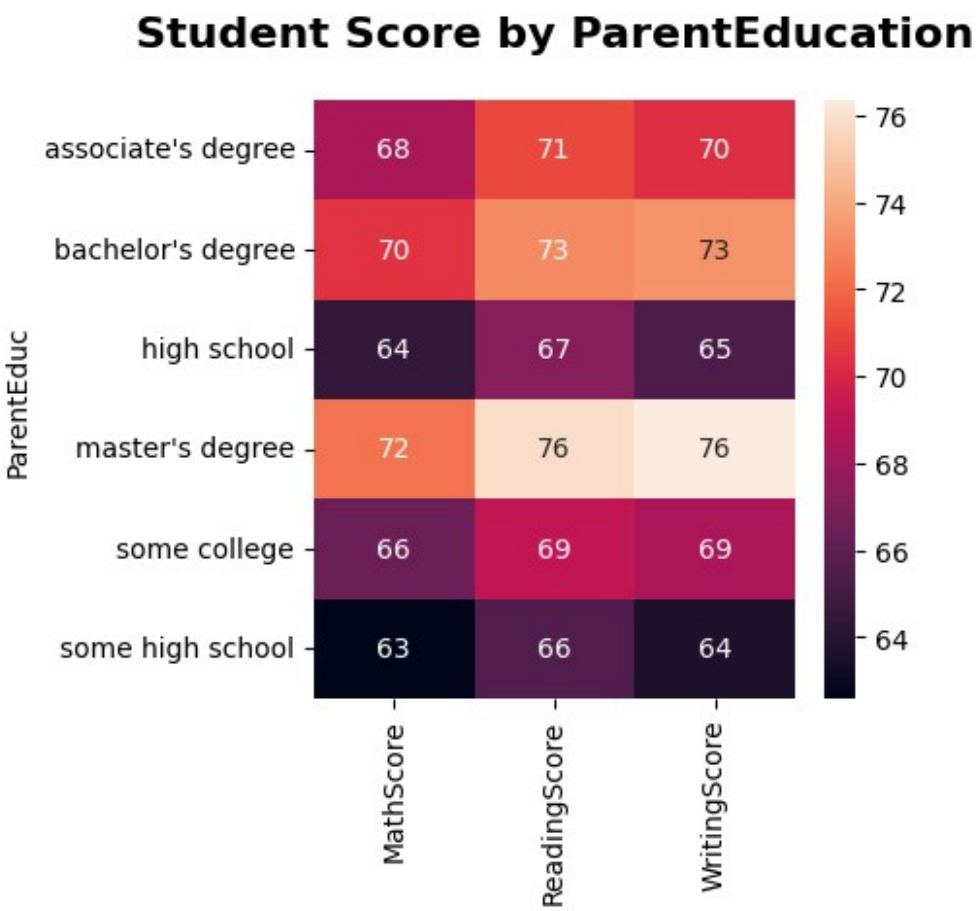



```
# From the above chart we have analysed that: the number of females in
the data is more than the number of males

gb = df.groupby("ParentEduc").agg({"MathScore": "mean", "ReadingScore"
: "mean" , "WritingScore": "mean"})

print(gb)
```

```
                     MathScore  ReadingScore  WritingScore
ParentEduc
associate's degree  68.365586     71.124324     70.299099
bachelor's degree   70.466627     73.062020     73.331069
high school         64.435731     67.213997     65.421136
master's degree     72.336134     75.832921     76.356896
some college        66.390472     69.179708     68.501432
some high school    62.584013     65.510785     63.632409
```

```
plt.figure(figsize=(4,4))
sns.heatmap(gb, annot= True ) # annot=True: Displays numerical values
in cells.
```

```
plt.title("Student Score by ParentEducation", fontweight='bold',
fontsize= 16 , pad= 20)

plt.show()
```

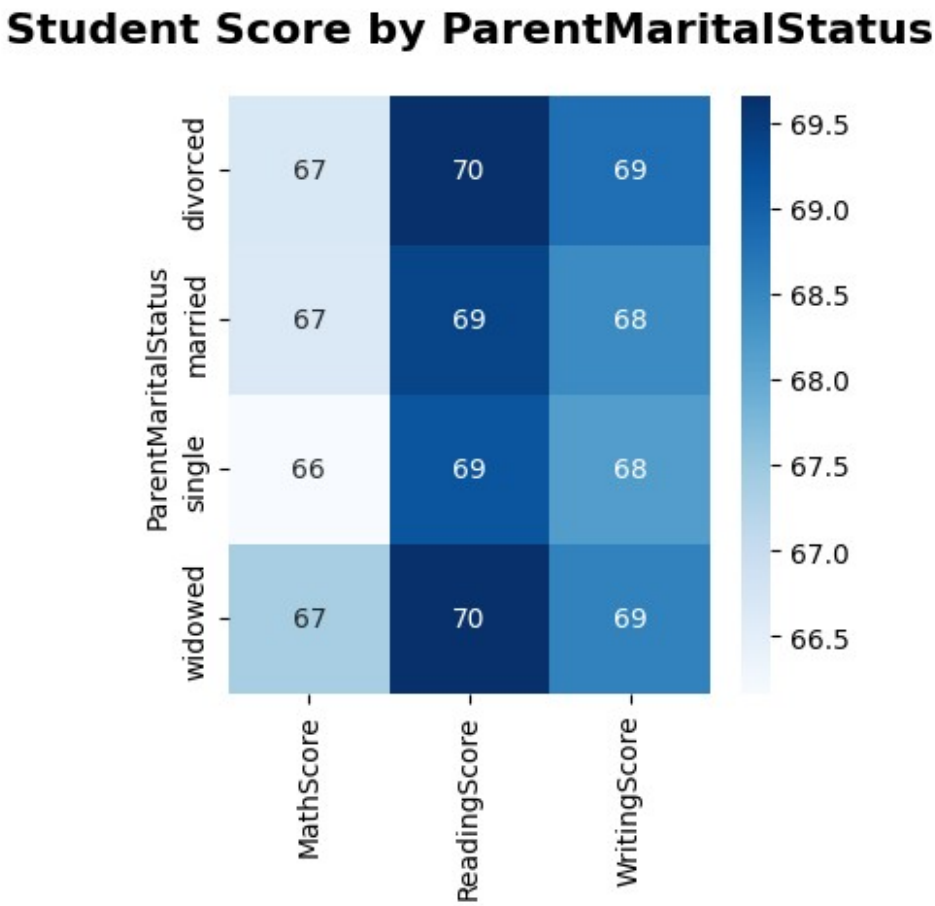



```
# From the above chart we have concluded that the education of the
parents have a good impact on the scores

gb1 = df.groupby("ParentMaritalStatus").agg({"MathScore": "mean",
"ReadingScore" : "mean" , "WritingScore": "mean"})

print(gb1)
```

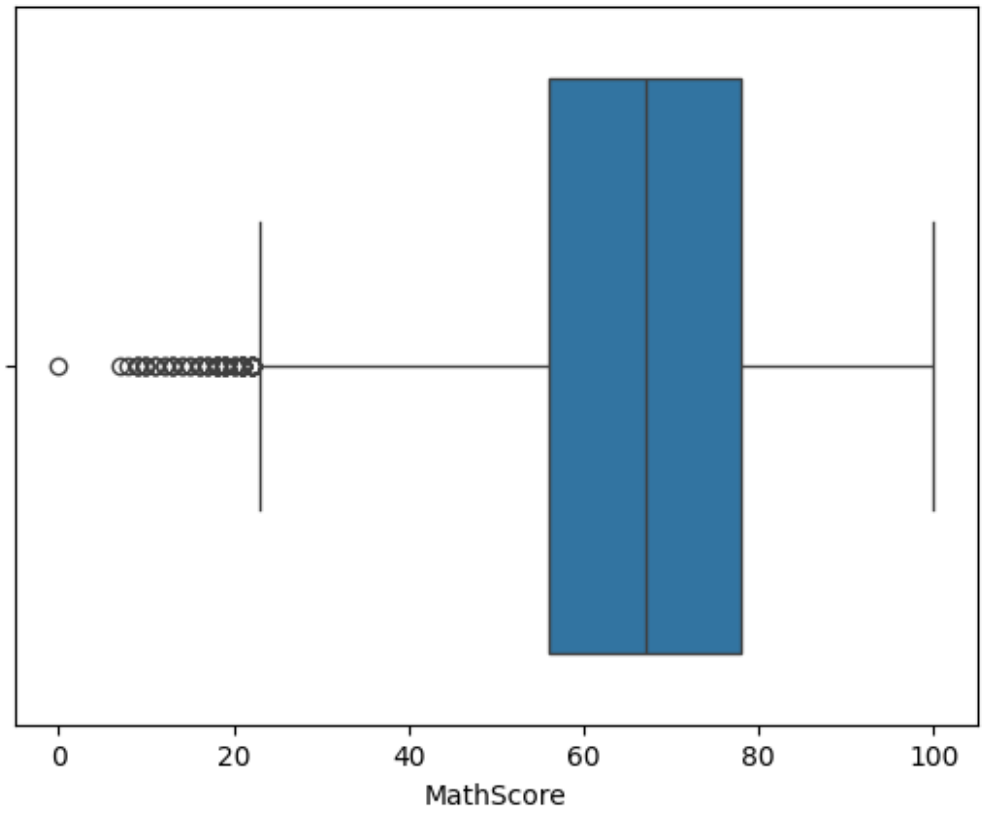
```
                      MathScore  ReadingScore  WritingScore
ParentMaritalStatus
divorced              66.691197     69.655011     68.799146
married               66.657326     69.389575     68.420981
single                66.165704     69.157250     68.174440
widowed               67.368866     69.651438     68.563452
```

```
plt.figure(figsize=(4,4))
sns.heatmap(gb1, annot= True ,cmap= "Blues" )
```

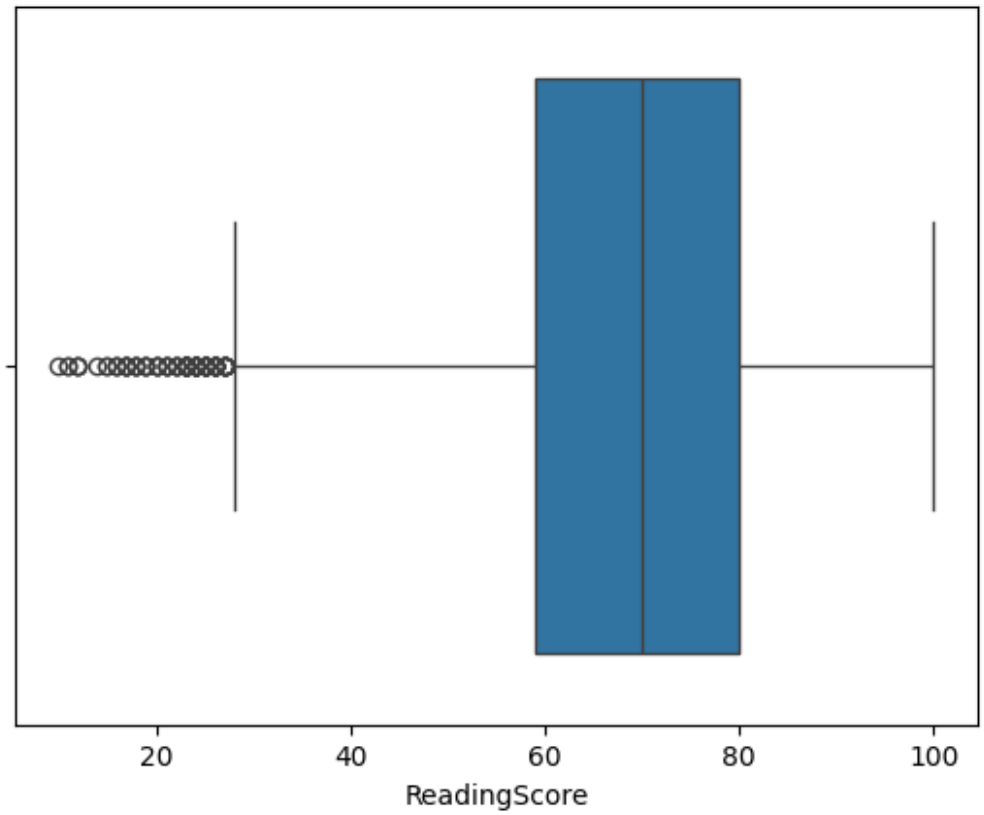
```
#annot=True: Displays numerical values in cells.

#colors customized using the cmap parameter (e.g., 'viridis', 'Blues',
'coolwarm').

plt.title("Student Score by ParentMaritalStatus", fontweight='bold',
fontsize= 16 , pad= 20)
plt.show()
```



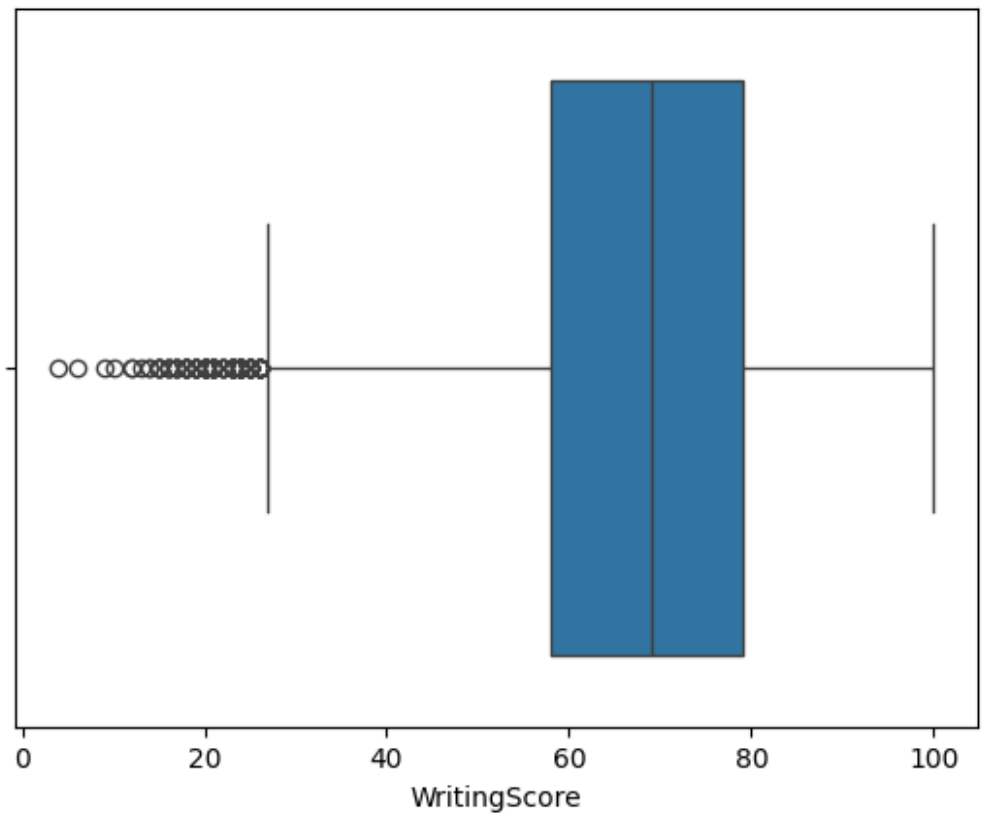

```
# From above chart we have concluded that there is no /negligible
impact on the student's score due to their parent's marital status

sns.boxplot(data= df , x="MathScore")
plt.show()
```




```
sns.boxplot(data= df , x="ReadingScore")
plt.show()
```




```
sns.boxplot(data= df , x="WritingScore")
plt.show()
```



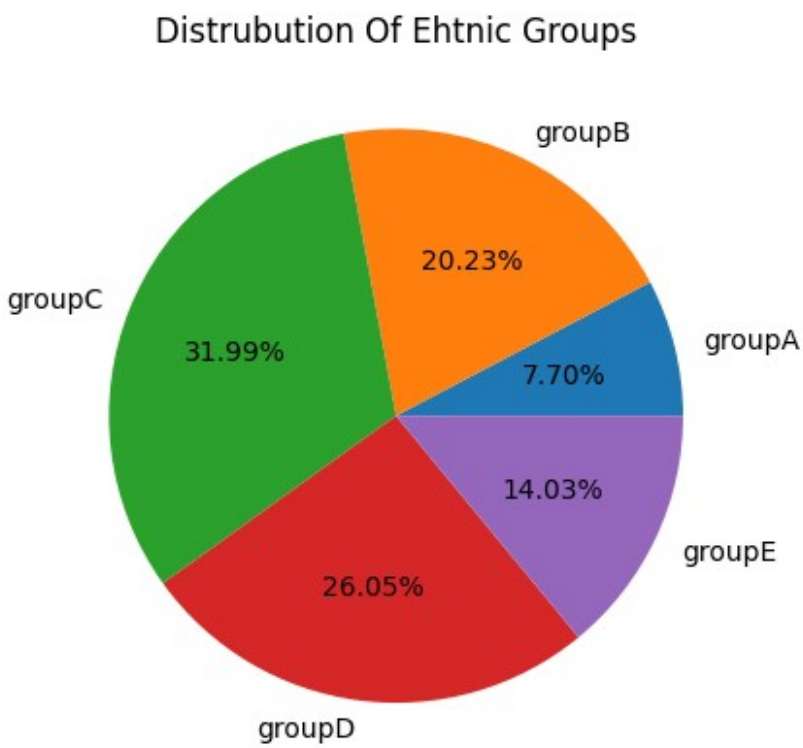

```
# From the above analysis of outliers we can concluded that studets
are weak at Maths compare to Read and Write

df["EthnicGroup"].unique()
```

```
<StringArray>
[nan, 'group C', 'group B', 'group A', 'group D', 'group E']
Length: 6, dtype: str
```

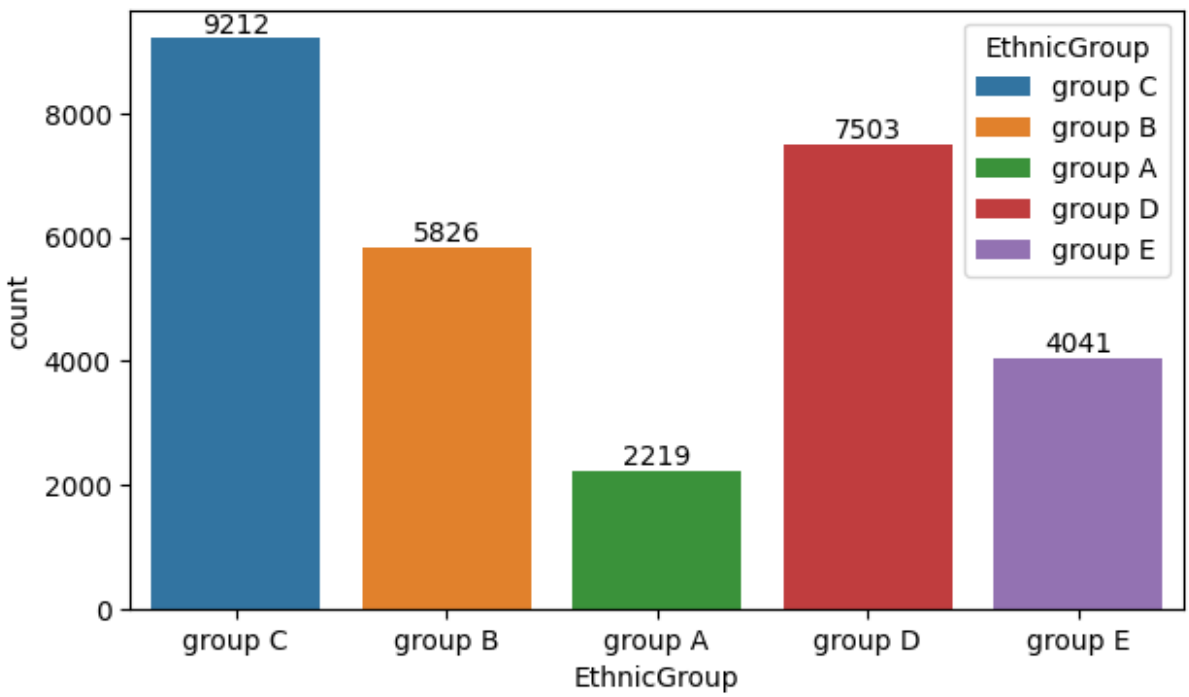
Distribution Of Ethnic Groups

```
groupA = df.loc[(df["EthnicGroup"] == "group A")].count()

groupB = df.loc[(df["EthnicGroup"] == "group B")].count()

groupC = df.loc[(df["EthnicGroup"] == "group C")].count()

groupD = df.loc[(df["EthnicGroup"] == "group D")].count()

groupE = df.loc[(df["EthnicGroup"] == "group E")].count()

l = ["groupA","groupB","groupC","groupD","groupE"]


my_list =[groupA["EthnicGroup"],
```

```
groupB["EthnicGroup"],groupC["EthnicGroup"],groupD["EthnicGroup"],grou
pE["EthnicGroup"]]

plt.pie(my_list, labels=l ,autopct="%1.2f%%")
#autopct is a parameter used specifically in Pie Charts (via
plt.pie()) to display the numerical percentage of each slice inside
the chart automatically

plt.title("Distrubution Of Ehtnic Groups")
plt.show()
```




```
plt.figure(figsize=(7,4))

ax = sns.countplot(data= df, x="EthnicGroup", hue='EthnicGroup')

sns.move_legend(ax, "upper right")

# Label all bars
for container in ax.containers:
    ax.bar_label(container)

plt.show()
```




```
# Conclusion : Group C has the highest number of records in the
dataset, followed by Group D and Group B.Group A has the lowest
representation, showing that the data is unevenly distributed across
ethnic groups.
```